

A Nanosyntactic account for the opaque morphology of Tibetic verbs

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Abstract. The verbal morphology of Classical Tibetan is notoriously complex, where a combination of concatenative and non-concatenative processes create paradigms of stem alternants with opaque many-to-many form-function mappings, where TAM features appear to be exponed by overlapping and discontinuous phonological material. This paper represents the first attempt at providing a generative, compositional analysis of the most morphophonologically complex Classical Tibetan verb paradigm (illustrated with $\sqrt{\text{kab}}$ ‘to cover’), and of its diachronic development into Themchen, a modern Amdo Tibetic variety. This analysis uses the Nanosyntax (NS) framework, where I argue that the lexicalisation algorithm can generate the paradigm’s overlapping and discontinuous exponence for free. Overall, this paper presents a case study for giving a compositional analysis to phenomena that appear to resist it, and the utility of compositional models that admit an intervening level of representation between form and function. By applying NS to data that it has not seen before, this paper also explores the consequences for prefix-suffix interaction within NS.

Key words: verbs, historical morphology, non-concatenative morphology, gestalt exponence, Tibetan, Nanosyntax

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Abbreviations

languages

CTib Classical Tibetan
ThCh modern Themchen Tibetan

Tibetic verb stems

PRS present in CTib
PST past stem in CTib
FUT future stem in CTib
IMP imperative stem in CTib and ThCh
IPV imperfective stem in ThCh
PFV perfective stem in ThCh

theoretical

NS Nanosyntax
DM Distributed Morphology
GE Gestalt Exponence
LI Lexical Item
FSEQ functional sequence
UR underlying representation
SR surface representation

1 Introduction

In this paper, I demonstrate Nanosyntax’s potential for generating opaque form-function mappings through the case study of one Tibetic verb: $\sqrt{\text{kab}}$ ‘to cover’. Its paradigm in Classical Tibetan and in modern Themchen Tibetan are presented in (1). $\sqrt{\text{kab}}$ exemplifies the most complex pattern of verb stem alternations in both Classical and modern Themchen Tibetan, and its Themchen forms only show the effect of regular sound changes, without undergoing any paradigm levelling or analogical extension in its diachronic development.

(1)

Classical Tibetan	IMP	PST	PRS	FUT
	<i>k^hobs</i>	<i>bkab(s)</i>	<i>ⁿgeb(s)</i>	<i>dgab</i>
Themchen	IMP	PFV	IPV	—
	[k ^h op]	[kwap]	[ŋgep]	—

As I will discuss in more detail in Section 2, the verbal morphology of historical Tibetan appear to resist morphemic decomposition, with morphophonological exponents often lacking correspondence to morphosyntactic featural classes. The possibility of advancing a decompositional analysis here thus becomes crucial to defending the universal applicability of morphological decomposition, upon which all morphology-as-syntax theories are based. I discuss this in Section 5, where I first compare the utility of paradigm-based vs decompositional morphological approaches for dealing with Tibetic verbs, and then defend the utility of a decompositional approach for the Tibetic data.

I then outline Nanosyntax, my chosen decompositional approach, in Section 4 and defend its utility for analysing morphological phenomena with opaque form-function mappings. My analysis is presented in Section 5, including synchronic analyses for the Classical and Themchen Tibetan paradigms for $\sqrt{\text{kab}}$, and a sketch of the diachronic development between them. The morphomic distribution of /-s/ vs onset voicing in $\sqrt{\text{kab}}$ ’s paradigm presents analytical difficulties, and I discuss the problem’s possible solutions in Section 6. This, however, does not diminish the analytical gains of my decompositional analysis for Tibetic verbs, since morphomic distributions are a difficult case for all morphological theories. Section 7 overviews overall successes and limitations of this paper.

2 Languages and phenomena

Tibetic languages are a subfamily within the Trans-Himalayan / Sino-Tibetan superfamily, and are spoken in the Tibetan plateau, its surroundings, and by Tibetan people in diaspora. Modern Tibetic varieties generally share morphosyntactic typology (incl. ergativity, complex evidentiality system) and lexis, while differing significantly in phonology (differing amount of complex syllable structure and tonogenesis) and degrees of mutual intelligibility (Tournadre & Suzuki, 2023). Classical Tibetan (CTib; 9-20th c. CE) is the historical literary language used in the Tibetan cultural region. It is highly standardised, diachronically stable, and archaising, preserving the morphophonology of Old Tibetan (7-9th c. CE; approximates Proto-Tibetic) or even a more distant ancestor of the language family before script-invention.¹

¹ Tournadre & Suzuki (2023) make the comparison between Tibetic languages and Romance languages: both exhibit a similar range of linguistic variation in the modern language family, while Classical Tibetan and Classical

2.1 Verb stem alternations in Classical Tibetan

The archaising orthography of Classical Tibetan (CTib) preserves complex morphological alternations in monosyllabic verb stems, which are the subject of this study. (2)² shows some examples of CTib verbs and their paradigms (morphological information from Hill, 2010),³ showing a range of variation for morphophonological processes, licit phonotactic structures, and patterns of syncretism.

(2)	IMP	PST	PRS	FUT
‘to be separated’	—	<i>bskal</i>	<i>bskal</i>	<i>bskal</i>
‘to chew’	<i>dmur</i>	<i>dmur</i>	<i>dmur</i>	<i>dmur</i>
‘to blame’	<i>ⁿphyas</i>	<i>ⁿphyas</i>	<i>ⁿphya</i>	<i>ⁿphya</i>
‘to chew’	<i>rmes</i>	<i>rmes</i>	<i>rme</i>	<i>rme</i>
‘to chop’	<i>tshogs</i>	<i>btsogs</i>	<i>ⁿtshog</i>	<i>gtsog</i>
‘to read’	<i>klogs</i>	<i>bklags</i>	<i>bklog</i>	<i>klog</i>
‘to pierce’	—	<i>btsugs</i>	<i>zug</i>	<i>zug</i>
‘to supplicate’	<i>khrold</i>	<i>bkral(d)</i>	<i>dgral</i>	<i>ⁿgrel</i>

The CTib verbal paradigm expresses a four-way tense/mood contrast. The four paradigm cells express imperative (IMP), past (PST), non-past (PRS for ‘present’), and future-oriented irrealis (FUT for ‘future’) Zeisler (2004:261).⁴ In my analysis, PRS and PST have realis/indicative mood,

Latin both persisted as highly standardised and diglossia-inducing cultural lingua francas. Unlike with medieval Romance varieties though, we do not seem to attest regional Tibetic varieties being written down alongside Classical Tibetan until the 20th century.

² In this paper I present Tibetic verb stems in a different order to how they are normally listed in Tibetic linguistics literature or in dictionaries. The standard ordering is PRESENT, PAST, FUTURE, IMPERATIVE for historical Tibetan, and IMPERFECTIVE, PERFECTIVE, IMPERATIVE for modern varieties. I use the latter order for the benefit of my Nanosyntactic stem tables in later sections.

Tibetan spellings are romanised using the Wylie system, except for the use of *ⁿ* for the Tibetan letter *a-chung* ཁྱ. *ⁿ* is conventionally transcribed using the apostrophe *'* in Wylie, and represents a nasal when occurring before another consonant (see Hill, 2019 for discussion).

³ Hill (2010) compiles verb information from nine existing dictionaries, eight of which were made by indigenous grammarians. These dictionary sources sometimes report different paradigms for the same verb. Where this has been the case for verbs I cite, I have chosen which forms to cite on a case-by-case basis. It is also possible that Hill (2010)’s sources report regularised / prescriptive inflectional paradigms rather than attested / descriptive ones. The latter issue is more difficult to mitigate without a large-scale philological investigation, so I content myself with merely acknowledging this confounding factor here.

⁴ ‘Imperative’, ‘past’, ‘present’, and ‘future’ are the conventional labels for the four CTib verb stems in the Tibetic linguistics literature, and I retain them here. My semantic interpretation of the four stems are based on Zeisler (2004)’s very detailed philological work on interpreting TAM in CTib. She uses very specific terminology for the functions of the four stems (incl. ‘non-past’ and ‘non-anterior’ for PRS, ‘purposive’ and ‘necessitative’ for FUT), and also identifies many overlaps between the functions of the four stems (e.g. all four stems can be used to express a prohibitive meaning), but the unique functions of each stem essentially boil down to ‘imperative’, ‘past’, ‘present’, and ‘future’. This interpretation of Tibetic verb stems is contra the more recent and equally painstaking work by Bialek (e.g. 2020, 2026), who argues that the four stems contrast primarily on aspect and voice rather than tense and mood. While aspects of Bialek’s argumentation are compelling, it is difficult to evaluate her claims rigorously (for example, she does not give clear diagnostics for what active and passive voice mean for Old/Classical Tibetan as an ergative language). Whether Zeisler’s system or Bialek’s is ultimately correct does not refute my core claims in this paper, which is that a framework like NS can be usefully used to analyse opaque

while FUT and IMP have irrealis moods. The interpretation that the ‘future stem’ gets its future-oriented semantics from being irrealis is supported by Zeisler’s report of CTib verb stem semantics, and the cotermination of future time reference and irrealis mood is also seen crosslinguistically (cf. e.g. Persson, 2008; Jendraschek, 2014). More nuanced or additional TAM distinctions than CTib IMP, PST, PRS, and FUT are expressed periphrastically with the help of auxiliaries, but those periphrastic auxiliaries must always select for one of the four primary stems.

As shown in (2), not all verbs have four distinct stems to fill the four paradigm slots. Some cannot fill all four slots due to semantic restrictions (non-volitional verbs cannot have imperative stems), and some verbs have the same syncretic stem across multiple paradigm slots.

Also evident from (2) is that there are many inflectional patterns for CTib verbs, which researchers have attempted to categorise into morphological classes. Notable recent attempts are Jacques (2012) and Hill (2010), both of which are based on the classification in Coblin (1976). Classifying CTib verbs into verb classes is a thorny problem because a) there are evidently many different inflectional patterns in the system (for verbs that can occupy all four paradigm slots, Coblin and Jacques’ schemes both find eight morphological classes, while Hill finds eleven); b) it is difficult to tease out the effects of analogy, alternative spellings, and prescriptive pressures, which create morphological irregularities that are ‘superficial’ to the underlying system. Successfully accounting for these multifarious verb classes will be crucial to any system-wide analysis of CTib verbal morphology, but is not of immediate concern to this paper since I am only analysing one verb class.

2.2 Verb stem alternations in Themchen Tibetan

Modern Themchen Tibetan (ThCh) is an Amdo Tibetic variety spoken in Themchen 天峻, located in Qinghai Province, China. Amdo is one of the most conservative groups in modern Tibetic languages, retaining complex syllable structures and the non-tonal phonology of Old/Classical Tibetan, and many of the historical verb stem alternation strategies discussed above,⁵ while these features have lost in the most innovative Tibetic varieties (e.g. Lhasa / Standard Tibetan). Within the conservative Amdo subgroup, Themchen is again one of the most phonologically and morphophonologically conservative (Bielmeier et al., 2018; Hua, 2002).

All modern Tibetic varieties that retain verb stem alternations have undergone paradigm restructuring since the CTib period. The FUT stem was lost in all modern Tibetic varieties (nuances see Tournadre & Suzuki, 2023:367); FUT is generally considered to have merged with the CTib PRS due to their stems being the most morphophonologically similar). The tense-based distinction between the CTib PRS and PST stems becomes an aspectual distinction between im-

morphological phenomena like Tibetic verb paradigms.

There is much work in Tibetic historical linguistics that seek to argue that the CTib inflectional paradigms I characterise as opaque in fact originated as transparent pieces of derivational morphology (e.g. Bialek, 2020). That work targets the reconstruction of an earlier stage of the language, whereas I am targeting the verbal morphology when it has grammaticalised into inflectional morphology.

⁵ This is at least true of Amdo varieties in the mid-late 1990s, when the fieldwork was conducted for Haller (2004) (my source of ThCh data for this paper), Hua (2002), and Bielmeier et al. (2018). Liu (2024)’s fieldwork on Labrang (conducted 2024), another Amdo variety that neighbours ThCh, finds that Labrang appears to have undergone rapid morphophonological change since its documentation in the late 1990s for Hua (2002).

perfective (IPV) and perfective (PFV) in modern varieties (Bielmeier et al., 2018). These developments in the paradigm functions of Tibetic verbs are summarised in (3). Additionally, all modern Tibetic varieties have developed complex evidential systems. Evidentiality, alongside other TAM information are expressed through auxiliaries which, like in CTib, select for one of the stem alternants. Again like in CTib, this paper is only concerned with the paradigm of stem alternants in ThCh.

(3)	CTib	IMP	PST	PRS	FUT
		↓	↓	↓	↓
	ThCh	IMP	PFV	IPV	—

(4) (morphological information from Haller, 2004) shows some examples of ThCh verbs and their paradigms, again showing a range of variation in phonotactic structures, morphophonological processes, and patterns of syncretism.

(4)		IMP	PFV	IPV
	‘to melt (intr)’	—	[ɕə]	[ɕə]
	‘to argue’	[ŋkʰon]	[ŋkʰon]	[ŋkʰon]
	‘to suck’	[ni]	[ni]	[nə]
	‘to chew’	[rdol]	[rdal]	[rdal]
	‘to send’	[ʂkər]	[ɸkər]	[ʂkər]
	‘to read aloud’	[tʰon]	[ton]	[ndon]
	‘to be caught on something’	[ŋgə]	[kwa]	—
	‘to lift’	[tɕʰoχ]	[ptɕaχ]	[ndzoχ]

3 Decomposition and opaque exponence

In CTib and ThCh verbs, exponence can involve a wide repertoire of morphophonological strategies (affixation, ablaut, voicing alternation, manner alternation). Most importantly for this study, exponence in these paradigms often involve what appear to be opaque many-to-many form-function mappings.

(5)		IMP	PST	PRS	FUT
	CTib				
	‘to read’	<i>klogs</i>	<i>bklags</i>	<i>bklog</i>	<i>klog</i>
	‘to supplicate’	<i>khrold</i>	<i>bkral(d)</i>	<i>dgral</i>	<i>ⁿgrel</i>
		IMP	PFV	IPV	—
	ThCh				
	‘to suck’	[ni]	[ni]	[nə]	—
	‘to chew’	[rdol]	[rdal]	[rdal]	—
	‘to lift’	[tɕʰoχ]	[ptɕaχ]	[ndzoχ]	—

This opacity is exemplified in (5), which repeats a few verbs from (2-4). CTib showing a more extreme version of the phenomenon because its phonotactics allow more complex syllable structures, but this is present in both CTib and ThCh. Sometimes exponence appears to be phonologically discontinuous with reference to function, as in CTib ‘to read’ *bklags*, where PST

seems to be expressed by *b-a-s*, or in ThCh ‘to lift’ [pt̪aχ], where PFV is associated with [p-a-]. Here it is also possible to say that, instead of *b-a-s* or [p-a-] constituting one morpheme, *b-*, *-a-*, and *-s* or [p-] and [-a-] are in fact separate morphemes. This breaks the apparent phonological discontinuity of these exponents, but raises the question of how the meaning of ‘past tense’ or ‘perfective aspect’ is distributed among these separate exponents. Their simultaneous presence distinguishes the CTib PST or ThCh PFV stems from other stems in their paradigms, but the presence of just one would not. For CTib *bklags*, *b-* and *-s* are both shared with another stem of the verb ‘to read’ (*b-* with PRS, *-s* with IMP), and while *-a-* is unique in the PST stem of ‘to read’, it is seen in both the PST and PRS stems of ‘to supplicate’. Similarly, while [-a-] is unique in ThCh PFV stem [pt̪aχ] for ‘to lift’, where the IMP and IPV stems share a vowel, the three ThCh verbs in (5) show that any two of the three stems can share a vowel in ThCh, and which two stems seems to be lexically determined.⁶ Some morphemic generalisations can be made about the verb forms in (2-4) (e.g. in both CTib and ThCh, PST or PFV stems are more likely to have a bilabial segment in the onset, and the vowel /a/), while the PRES or IPV stems are associated with prenasalisation), but there are many more instances where form and function are not mapped transparently in these paradigms.

Overall, the effect is that exponence is often opaque and feature many-to-many form-function mappings in Tibetic verbs, because **a**) either exponence is discontinuous with respect to a stem’s function or its function is scattered across multiple smaller exponents that must jointly expone the stem’s function, and **b**) the individual small exponents are often shared between different stems in a way that does not correspond to natural classes in stem function.

This opacity in Tibetic verbal morphology exemplify what has been variously termed ‘reciprocal conditioning’ (Anderson, 1992:55-56), ‘extended exponence’ (Matthews, 1991:180ff.), ‘gestalt exponence’ (Blevins, 2016:73ff. in this article I use ‘gestalt exponence’ or GE to refer to this phenomenon) in the Word and Paradigm morphology (WP) literature. GE describes cases of exponence where the same morphophonological material appears in multiple different contexts, and the same morphosyntactic function is expressed by multiple pieces of morphophonological material, none of which are unique to that function – which is precisely the kind of form-function opacity discussed for (5) above. For WP, GE demonstrates the non-utility of morphological decomposition: the whole / ‘gestalt’ surface form is parsable through comparison with other surface forms in the paradigm, but that surface form cannot be split into discrete parts that represent stable components of the form’s overall meaning.

Though GE has been useful for defending WP, WP has only been partially useful for explicating GE. More recent quantitative WP work on the mathematical complexity of morphological paradigms explore the degrees to which morphological paradigms are learnable and diachronically (un)stable (e.g. Ackerman & Malouf, 2013; Round, 2026), which offer accounts of why GE and other apparently very complex paradigms are learnable and diachronically (un)stable. Liu

⁶ An alternative account to ThCh vowel-sharing being ‘lexically determined’ is that it is systematically determined by morphological class, the three verbs given here fall into different morphological classes. However, vowel-sharing crosscuts other morphological faultlines like affixation, semantic categories, and diachronic pathways in ThCh verbs, meaning that it does not seem advantageous to base verb conjugations on vowel-sharing.

(2024) finds that even though the CTib verb paradigm looks to be made up of many interacting exponents and multitudinous inflectional classes, its informational complexity ('conditional entropy' after Ackerman & Malouf, 2013) does not exceed that of simpler-looking morphological systems. Additionally, the CTib verb stem that was lost diachronically (the future stem) had a higher conditional entropy than the other stems in CTib, consistent with Ackerman & Malouf (2013)'s prediction that the most informationally unpredictable part of a morphological paradigm is most prone to erosion. Findings like these answer the question 'how do we generate GE paradigms' from cognitive and diachronic perspectives, but do not offer any insights on that question posed from a formal theoretical standpoint. Answering this latter question thus falls to decompositional generative approaches.

The challenge that GE poses towards discrete morphological decomposition makes it a natural sibling of non-concatenative morphology (incl. templatic morphology, reduplication, subtractive or augmentative processes, sound alternations). Both challenge the 'classical' notion of decompositionality (or what Bye & Svenonius, 2012 term the 'concatenative ideal'), where morphology is decompositional if wordforms are composed of discrete sub-word form-function mappings. Non-concatenative morphology contravenes the phonological element of this concatenative ideal, while GE its syntactic/semantic one. As shown in Tibetic verbal paradigms, GE can involve phenomena that are phonologically concatenative (segmental affixation) or non-concatenative (segmental and featural alternations), but their many-to-many form-function mappings violate what the contiguity and autonomy of idealised morphemes as framed by Bye & Svenonius (2012), respectively that morphemes cannot be discontinuous, and that they should have stable functional contents.⁷

The strategy for rehabilitating non-concatenative morphology in decompositional approaches is to introduce abstractness into the phonological representations and processes, so that the underlying representation can be analysed concatenatively, while phonological operations introduce non-concatenative effects into the surface form (e.g. Bye & Svenonius, 2012; Davis & Tsujimura, 2015). Therefore it seems natural to attempt a similar approach for creating decompositional analyses of GE. If GE involves opaque many-to-many form-function mappings, then we may be able to generate such mappings in a framework where there is (so to speak) 'more going on' in the system that maps function to form, such that the mapping process is consistent and constrained, but generate a complex mapping that appears opaque at first glance. As I will demonstrate in the next section, Nanosyntax has the potential to do exactly this.

⁷ Bye & Svenonius (2012)'s original formulation of morphemic 'autonomy' only addresses the phonological dimension, i.e. the idealised morpheme should have one stable surface form. This befits the fact that their paper focuses on resolving the phonological effects around non-concatenative morphology, but their overarching sentiment that the idealised morpheme is stable and context-free can be straightforwardly and relevantly extended to the functional aspect of morphemes.

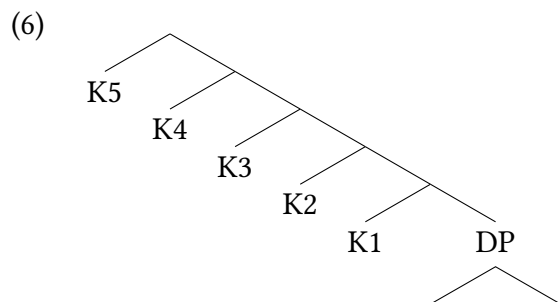
4 The framework: Nanosyntax

Nanosyntax (NS) is a formal generative approach to morphology that originated with Michal Starke’s work in the 2000s, and De Clercq et al. (2025) give a more recent overview of the framework. NS uses standard generative syntactic stipulations like the Y-model of the grammar, binary-branching hierarchical structure, and functional projections into the sub-word domain, and states that phonological forms are mapped to branching hierarchical structures, which are themselves derived from constrained syntactic movements. NS differs from Distributed Morphology (DM, the other leading generative framework for morphology) in that, in DM, spellout targets bundles of functional features positioned at syntactic terminals, whereas in NS, every functional feature has its own syntactic terminal, so spellout (often called ‘lexicalisation’ in NS) targets specific hierarchical relationships between features as well as the features themselves. This means that NS effectively injects an extra layer of representation between the ‘form’ (phonological representation) and ‘function’ (functional features) of each lexical item, in which reside the mappings between form and function (the hierarchical relationships between features). Different hierarchical relationships between features is enough to produce different spellout outcomes even with the same set of functional features, and the lexicalisation algorithm sometimes forces the same feature to surface multiple times as a part of different LIs. These properties of NS mean that, even though the framework was originally built for analysing non-GE morphological phenomena, its native apparatus is able to generate opaque form-function mappings ‘for free’.

I now introduce the relevant key tenets of NS, and its lexicalisation algorithm. These are articulated in more detail in the studies collected in Caha et al. (2025), and in the works cited there.

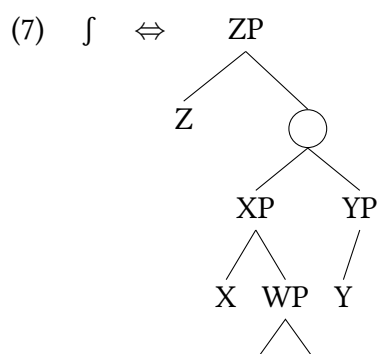
Functional sequence (FSEQ). NS takes a cartographic approach towards functional features (after Cinque, 1999), namely that functional features are drawn from a universal set and are merged into the derivation following an universal order.

Featural containment. Features are semantically abstract in NS, so that (for example) case features may well be called K1, K2, etc as in (6) rather than Nominative, Accusative, etc. Within the same domain (e.g. the case domain in (6)) features have a containment relationship, i.e. having a higher feature (e.g. K4) entails having all of the lower features in that domain (K1-3). Synsem contrasts are thus not generated by having a specific feature that ‘means’ accusative or instrumental, but by how many features a given wordform has from the FSEQ. Having more features in a given domain means that the wordform is more semantically specified in that domain, and vice versa. So in (6), not having K1 means that the DP cannot have any case interpretation, having only K1 gives the default case (nominative), having K1 and K2 gives accusative, K1-3 genitive, K1-4 dative, K1-5 instrumental (after Caha, 2009).



Late Insertion. Like other syntactic approaches to morphology that are based on the Y-model of the grammar, the syntactic part of a lexical item (relevant functional features and their syntactic relation to each other) is stored separately from its phonological part, and the syntactic structure is built first before the insertion of phonological material.

Phrasal lexicalisation, the Superset Principle. Lexical items (LIs) map phonological to syntactic material, and a syntactic structure can be lexicalised when there is an LI whose syntactic part matches it. In NS, an LI can also lexicalise a structure when that structure is a subset of the structure specified in that LI ('Superset Principle'), provided that the subset structure is a phrasal constituent ('phrasal lexicalisation'; this is in contrast to 'span-based' approaches to NS, e.g. Svenonius, 2012). For example, the sample NS LI in (7) shows the LI for /f/ and its corresponding syntactic structure. /f/ can be inserted if there is a constituent in the syntactic structure that is an exact match for ZP in (7), or if there is a constituent that exactly matches one of the constituent phrasal structures within ZP – i.e. the circle node⁸, XP, WP, or YP. /f/ cannot lexicalise just [X], [Y], or [Z] because those are syntactic terminals and not phrases, and it also cannot lexicalise [Z, X] or [WP, Y] (for example) because those do not form a constituent.



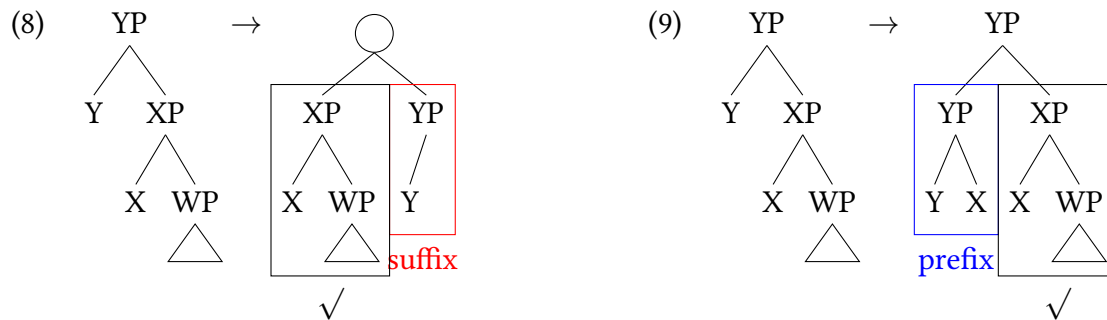
Late Insertion, lexicalisation-driven movements. NS is a Late Insertion framework in that a syntactic structure is built first before phonological material that matches it is inserted. In NS, lexicalisation (i.e. the insertion of phonological material) must occur each time a new feature is merged, and only when lexicalisation is successful (i.e. every constituent within the derivation

⁸ This node is unlabelled, but marked with a circle for ease of reference. This unlabelled node, as well as YP being unary branching in (7), both arise from lexicalisation-driven movement (see main text below). Letting LIs reference the result of movement originates with Blix (2021).

matches some LI or a subset of an LI) can the next feature in the fSEQ be merged. If, when a feature is merged, the resulting structure cannot be lexicalised, an algorithm kicks in where the derivation tries different movement options to see if one of their outcomes can be successfully lexicalised. The algorithm terminates when lexicalisation is successful, and the derivation can proceed into a new cycle where the next feature is merged. Thus, syntactic operations (merge and move) are overall interleaved with lexicalisation into phonological forms, but NS is still a Late Insertion framework because syntax strictly proceeds lexical insertion within each merge cycle.

Cyclic overwrite. Even though NS requires successful lexicalisation in each cycle to proceed to the next, in the new cycle, the existing lexicalisation is ‘forgotten’ and the lexicalisation attempt starts from scratch. This means that the lexicalisation attempts in successive cycles can potentially look very different from each other, if the merging of one new feature means that the structure is lexicalised by a different set of LIs.

Linearisation is determined by movement, prefixes and suffixes arise from different syntactic structures. NS has strong antisymmetric commitments (following Kayne, 1994), using hierarchical structure to generate linear order. New features are merged into head position on a left branch by default (Y in the first tree in (8) and (9)). Suffixes are generated from the root (XP) moving into spec position ‘around’ the feature (Y), as in (8). Suffixes are unary branching at the bottom because they are always remnant from the root undergoing comp-to-spec evacuation. Prefixes are generated from building a new subtree with the feature in the current cycle (Y) and the feature from the previous cycle (X) as in (9) (Starke, 2018). They are always binary-branching at the bottom because they are not remnants from movement.



The lexicalisation algorithm. When a feature is merged, we attempt to lexicalise every phrasal constituent in the derivation. The structure cannot be fully lexicalised, a lexicalisation algorithm is initiated where the derivation tries different repair strategies until the structure can lexicalise. There have been different versions of this algorithm in the history of NS (e.g. Caha, 2009; Starke, 2018), but I follow here Caha’s 2023 algorithm as articulated in De Clercq et al. (2025), with the prefix- / Complex Left Branch formation algorithm of Starke (2018). The steps of the lexicalisation algorithm are illustrated in (11), with a) as the starting point when a new feature (Z) is merged.

(10) The NS lexicalisation algorithm:

1. Move the highest labelled non-remnant node into Spec position, creating an unlabelled mother node. If the result still cannot lexicalise, try option 2.

In (11), this is XP, and this movement step is shown in b). Without XP, the only daughter of the unlabelled circle node is YP, so the circle node is ‘telescoped’ up and ZP ‘inherits’ YP as a daughter.

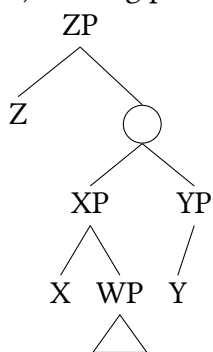
2. Undo the movement from the previous step, and move the mother node of the node that was moved last time. If the result cannot lexicalise, try this step recursively until we run out of eligible nodes. If we still cannot lexicalise, try option 3.

In (11), the only eligible node for this step is the circle node, and its movement is shown in c).

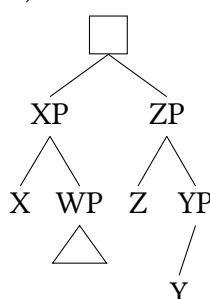
3. Undo any movements from the previous step, and open a new workspace (i.e. tree). In this new workspace, merge the feature of this current cycle (head) with the feature that was successfully merged in the last cycle (comp).⁹ Merge this into the existing (‘main-line’) derivation as a Complex Left Branch (CLB). The head in the CLB can project into the main spine.

In (11), Y and X form a CLB as in d).

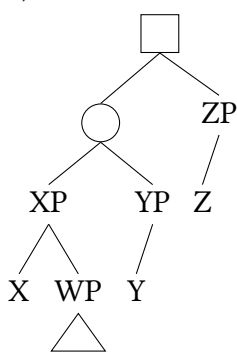
(11) a) starting position: merge Z



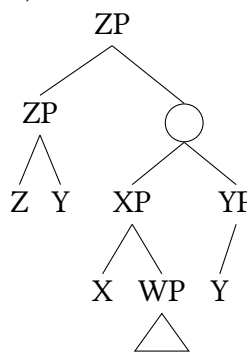
b) move labelled non-remnant



c) move the mother node



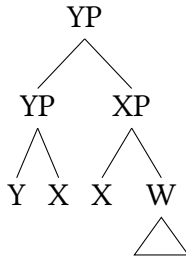
d) form a CLB



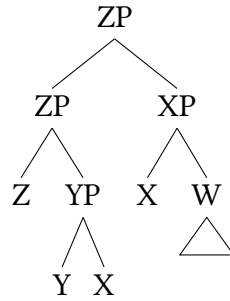
⁹ More recent work (e.g. presentations at the 2026 DM Meets Nano Workshop) is moving away from using the last-merged feature in comp-position when forming a CLB, preferring a placeholder object (notated .) instead. However, for now I stick with the older algorithm that uses a contentful feature in comp since it creates the effect of one feature being exponed in multiple positions in a derivation.

When a new feature is merged into a derivation where there already is a CLB (12a), the feature is tried inside that CLB first (12b),¹⁰ and only when it cannot lexicalise within the CLB, a new CLB is formed (12c) with the new feature (Z) in head position, and the last feature that was successfully merged in the mainline derivation (X) in comp position.¹¹

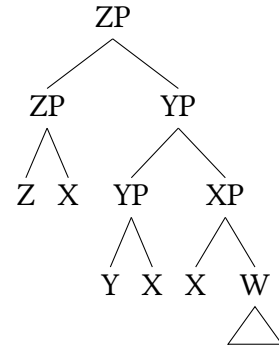
(12) a) starting point



b) existing CLB



c) new CLB



Many versions of the NS algorithm include an option for backtracking, where if the derivation cannot lexicalise, the derivation retreats to the previous cycle (or the previous juncture where an untried movement option remained), tries the next option in the algorithm, and rebuilds the derivation from there (De Clercq et al., 2025). Additionally, backtracking traditionally follows lexicalisation-driven movements but precedes the creation of CLBs (Starke (2018) in fact proposes CLBs as a final option when backtracking fails). There are many incontrovertible issues with backtracking. It not only drastically complicates the decision tree for building a given structure (Blix, 2021), but also builds lookahead into the system, which is traditionally undesirable in generative theory. Additionally, I have not found an argument for why backtracking should not be infinite. Algorithms that include backtracking limit it to undoing one choice in the lexicalisation algorithm per merge-cycle (e.g. Starke, 2018; Caha et al., 2025), but this seems to be an arbitrary decision to limit backtracking's power rather than something inherent or necessary to the mechanism. The decision for backtracking to precede CLB-/prefix-formation also appears to be an arbitrary one, deriving from the fact that CLBs were developed later than backtracking. I have not included backtracking as part of my algorithm above, because I believe backtracking should be eliminated where possible, and minimised if elimination is impossible. Where backtracking is required for a derivation to work, it should be the last step in the algorithm (i.e. *following* prefix-/CLB-formation), and perhaps allowed to backtrack multiple cycles at a time. The main analysis in this paper does not use backtracking.

¹⁰ I assume that when a new feature is merged into a CLB-having structure, it is first merged at the very top of the structure (i.e. this feature is head, all of the existing derivation is comp) and lexicalisation-driven movements are tried. When it would ordinarily be time to create a CLB, the new feature is first tried as the head of the existing CLB. There are different interpretations to whether, at this point, the new feature is also tried in the head position of the existing 'mainline' right branch (discussed by Demonie, 2025 as 'Multiple Merge, after Caha, 2019), but the choice here does not change anything substantive about my present analysis.

¹¹ I have not so far found justification for why the second CLB must use the last feature from the mainline derivation (X in (12)) rather than the last feature that was merged overall (Y), but I follow Wiland (2025) in this, who is interpreting Starke (2018).

5 Nanosyntactic analysis for the paradigm of $\sqrt{\text{kab}}$

5.1 Phonological decomposition

In (13), the CTib surface forms of $\sqrt{\text{kab}}$ are analysed into concatenative morphological sequences.

(13)		IMP	PST	PRS	FUT
	SR	<i>k^hobs</i>	<i>bkab(s)</i>	<i>ⁿgeb(s)</i>	<i>dgab</i>
	a) interpret optional affixes	—	bkabs	ⁿ geb	—
	b) aspiration is allophonic	kobs	—	—	—
	c) identify concatenative affixes	kob-s	b-kab-s	n-geb	d-gab
	d) voicing alternation as affix	—	—	n-V-keb	d-V-kab
	e) vowel ablaut as affix	o-kab-s	—	n-V-kab	—
	UR	o-kab-s	b-kab-s	n-V-kab	d-V-kab

To comment additionally on some of the steps in (13): In step (13a), both the PST and PRS stems are associated with -s, but because -s is overall more associated with the PST stem than PRS (Hill, 2010), I retain -s for PST and not for PRS. For (13b), Hill (2007) finds that voiceless obstruents are aspirated syllable-initially or after ⁿ. For (13d), I analyse voiced syllable onset g as k with a voicing prefix /V-/, adjacent to the segment undergoing voicing alternation. The segment undergoing voicing alternation as part of verbal morphology is always an obstruent, and always the first segment in the verb $\sqrt{\text{root}}$ (Hill, 2010). For (13e), I take -a- (which appears in two stems, PST and FUT) as the underlying root vowel. Since in both CTib -e- and ThCh [e] only co-occurs with the nasal prefix, I treat it as an effect of /n-/. I analyse -o- as its own morpheme since it is not possible to reduce it to the effect of another morpheme, and write it as a prefix /o-/, which will overwrite the root vowel in the phonology.

(14) shows the phonological decomposition of the ThCh surface forms for $\sqrt{\text{kab}}$ into underlying representations, taking care to align with the URs in (13) when possible.

(14)		IMP	PFV	IPV
	SR	[k ^h op]	[kwap]	[ŋgep]
	a) aspiration is allophonic	kop	—	—
	b) undo final devoicing	kob	kwab	ŋgeb
	c) affixes for ablaut and voicing alternation	o-kab	—	n-V-kab
	d) [-w-] as the result of bilabial prefix	—	b-kab	—
	UR	o-kab	b-kab	n-V-kab

Steps (14a) and (14c) are the same as their analogous steps in (13) above. For (14b), because final obstruent voicing is not contrastive in ThCh (Haller, 2004), we can analyse [-p] as the result of final devoicing, bringing the ThCh form closer to final /-b/ from the CTib UR. For (14d), we can view [kw-] as the realisation of a [+bilab]- prefix with root onset /k-/. The ThCh PFV stem where [+bilab]- appears is the reflex of the CTib PST stem, and just as b- was overwhelmingly associated with CTib PST stems, bilabial onsets commonly occur in ThCh PFV stems. These bi-

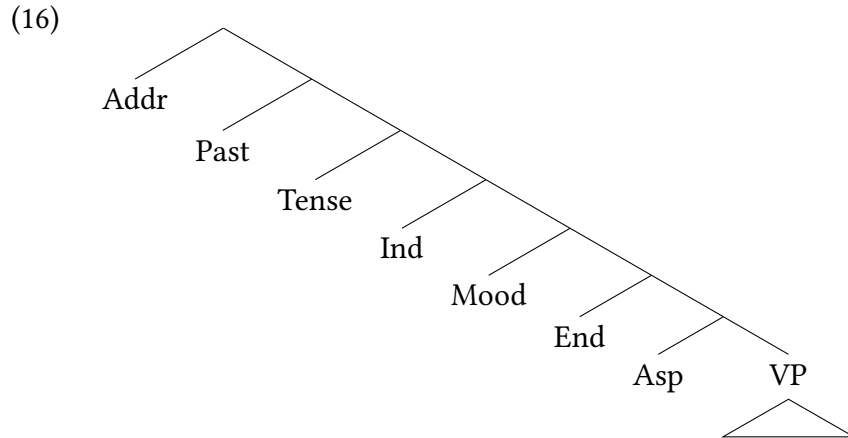
labial onsets take a variety of phonetic forms, as seen in (15), which shows example verbs that have a velar onset consonant in their root (morphological information from Haller, 2004). The continuity between [p-], [p-w-], and [-w-] suggests that all three *are* likely reflexes of CTib *b*-, and their lexically-conditioned co-existence perhaps suggests ongoing change in the morphology of the PFV stem at the time the data was collected. Thus I think it overall justified to posit that, in (14), [kw-] is the realisation of a bilabial prefix with /k-/, and this can be represented as /b-k-/ in (14d).

(15)

	IMP	PFV	IPV	realisation of [+bilab] in PFV
‘to cover’	[k ^h op]	[kwap]	[ŋgep]	[-w-]
‘to spin (e.g. wool)’	[k ^h u]	[kwa]	[ŋk ^h i]	[-w-]
‘to laugh’	[rgol]	[bgwal]	[rgol]	[b-w-]
‘to bend’ (tr.)	[k ^h əç]	[pkəç]	[ŋgəç]	[p-]

5.2 Syntactic/semantic decomposition

(16) shows my FSEQ for this analysis.



VP contains lower projections such as ones for lexical semantics (features like Cause or Init for casuation, Proc for change of state, etc; e.g. Cortiula, 2023) and category information. Above VP, features for aspect, mood, tense, and discourse features are merged in that order (Cortiula, 2023).¹² The first feature in the aspectual domain is called Asp(ect), the first feature in the mood domain is called Mood, the first feature in the tense domain is called Tense. Each domain has one additional feature (here called End, Ind, and Past), meaning that there is a possible two-way distinction in each domain, where having both features means a more specified semantics than only having the first feature (as per ‘featural containment’ in Section 4). Having Asp only is imperfective, Asp and End is perfective (Wiland, 2025). Having Mood only is irrealis, Mood and Ind is realis/indicative (Starke & Cortiula, 2021). Having Tense only is non-past, Tense and

¹² This is a different ordering than from Cinque (1999), where mood, tense, and aspect nodes are interspersed, but mood nodes are generally above tense and aspect nodes. However, the ordering of mood > tense > aspect appears standard in recent NS work (Caha et al., 2025; Cortiula, 2023).

Past is past (Cortiula, 2023). After Demonie & De Clercq (2023), the imperative is analysed as containing an Addr(essee) feature, merged high because it references the discourse.

(17) restates the verb stems in CTib and ThCh and their diachronic correspondence.

(17)	CTib	IMP	PST	PRS	FUT
		↓	↓	↓	↓
	ThCh	IMP	PFV	IPV	–

(18) shows my analysis for the features present in each CTib and ThCh verb stem (lowest feature on the left).

(18)	stem	VP	Asp	End	Mood	Ind	Tense	Past	Addr
	CTib								
	IMP	✓	–	–	✓	–	–	–	✓
	PST	✓	✓	–	✓	✓	✓	✓	–
	PRS	✓	✓	–	✓	✓	✓	–	–
	FUT	✓	✓	–	✓	–	✓	–	–
	ThCh								
	IMP	✓	–	–	✓	–	–	–	✓
	PFV	✓	✓	✓	✓	✓	–	–	–
	IPV	✓	✓	–	✓	✓	–	–	–

The IMP and CTib FUT stems are irrealis through having only Mood and no Ind, and distinguished from each other by FUT having a temporal domain (Tense for ‘non-past’) and IMP lacking it. CTib tense and ThCh aspect are distinguished through containment relationships within those domains. The CTib paradigm does not have a synsem contrast in the aspectual domain because its stems either lack the aspectual domain altogether (IMP, with no aspectual features), or have the same single Asp feature (the other three stems), and this is able to develop into the aspect-based ThCh paradigm by having one stem (CTib PST → ThCh PFV) gain an additional aspectual feature (End). When the CTib tense distinction is lost, however, ThCh loses the tense domain completely, to account for the fact that aspectual stems can be used in temporal context.

5.3 Mapping forms to features for $\sqrt{\text{kab}}$

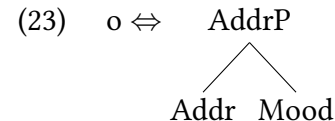
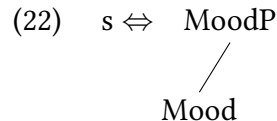
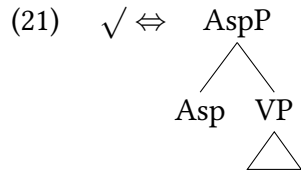
The phonological and syntactic decompositions for each stem of $\sqrt{\text{kab}}$ are synthesised in (19) and (20), which shows each exponent in the CTib and ThCh paradigms for $\sqrt{\text{kab}}$ and what features they each lexicalise.

(19)	SR	UR	stem	VP	Asp	End	Mood	Ind	Tense	Past	Addr
a)	<i>k^hobs</i>	o-kab-s	IMP	kab			-s				o-
b)	<i>bkab(s)</i>	b-kab-s	PST	kab			-s		b-		
c)	<i>ⁿgeb(s)</i>	n-V-kab	PRS	kab			V-		n-		
d)	<i>dgab</i>	d-V-kab	FUT	kab			V-		d-		

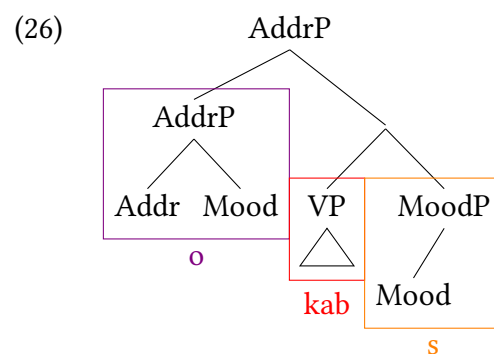
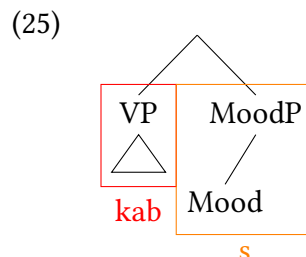
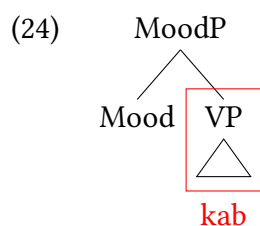
(20)

	SR	UR	stem	VP	Asp	End	Mood	Ind	Tense	Past	Addr
a)	[k ^h op]	o-kab	IMP	kab			o-				o-
b)	[kwap]	b-kab	PFV	kab			b-				
c)	[ŋgep]	n-V-kab	IPV	kab			V-	n-			

As an example, I show the derivation for the CTib and ThCh IMP stems, starting with CTib /o-kab-s/. This form uses the three LIs in (21-23).

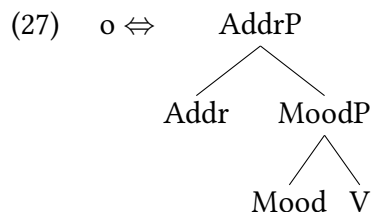


Assuming the successful merge and lexicalisation of lower projections, we begin with the formation of VP. This is a subset of the $\checkmark$ LI, so it is successfully lexicalised as the verb root. We then merge the next feature in (19a), Mood, into head position (24). This cannot successfully lexicalise, so the lexicalisation algorithm moves the closest labelled non-remnant into spec position (step 1 in (10)), obtaining the structure in (25), which can now be lexicalised by /kab/ and /-s/.¹³ When the next feature for IMP, Addr, is merged, none of the lexicalisation movements in (10) can produce a lexicalisable outcome. So, Addr creates a new workspace with Addr as head and Mood (the last successfully merged feature) in comp, and this AddrP can be lexicalised by (23). This workspace is merged into the mainline derivation as a CLB, and projects AddrP over all (26).

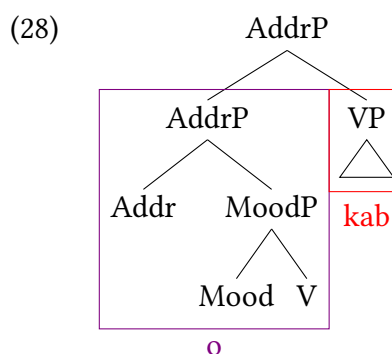
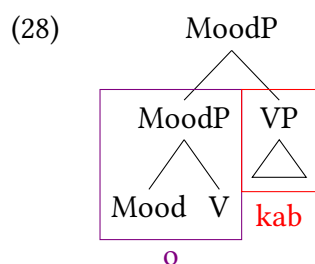


ThCh uses the same $\checkmark$ LI, but one /o-/ LI (27) for the lexicalisation cycles for both Mood and Addr.

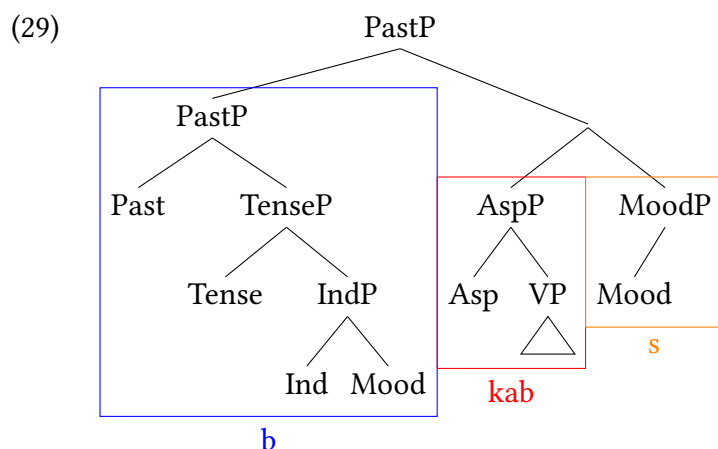
¹³ In NS, because linearisation is determined by hierarchical positioning, LIs don't inherently specify whether an affix is a prefix or suffix. This means that phonological forms of LIs are /V/, /s/, /o/ etc as shown in (30-41). However I often refer to affixes as /V-/, /-s/, /o-/ etc in prose and in other tables, to remind readers of where these forms are supposed to surface relative to the root.



The derivation begins identically as CTib to (24), but the movement in (25) cannot save it because the /-s/ LI has been lost in ThCh. Therefore, a CLB is created straightaway with Mood and V (28). When Addr is then merged, it can successfully lexicalise in its original head position in the CLB, thereby producing the final structure in (28).

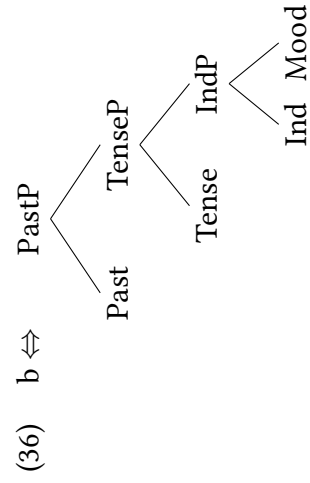
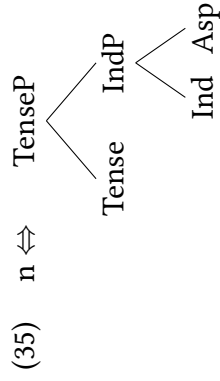
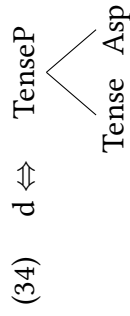
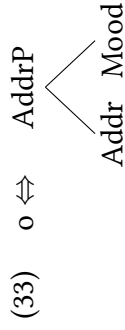
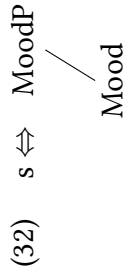
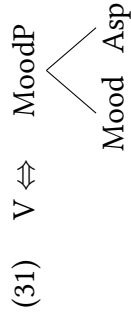
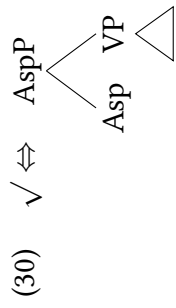


As another example, (29) shows the final derivation for the CTib PST stem. After the Mood cycle, the overall resulting structure is the unlabelled node. Next, Ind is merged, and since there is no way to lexicalise using either of the movement options, a CLB is opened with Ind as head and Mood (the last successfully merged feature) as comp. This CLB is lexicalised as /n-/ initially, because it is a subset of both /n-/ and /b-/, but /n-/ is preferred because it is the smaller LI and less material is ‘wasted’.

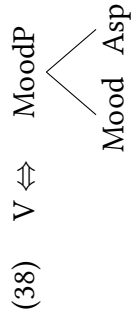
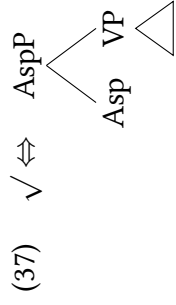


The entire paradigms for $\sqrt{\text{kab}}$ can be derived using the LIs in (30-41). The CTib LI for /V-/ (31) deserves further comment, discussed in Section 6.

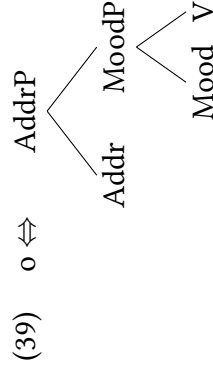
LIs for CTib derivation



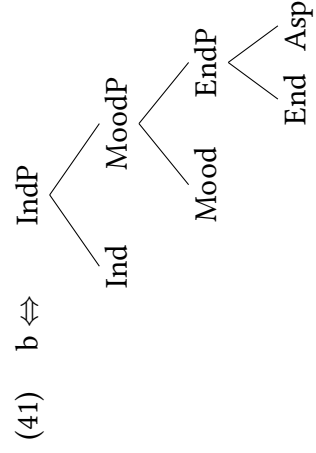
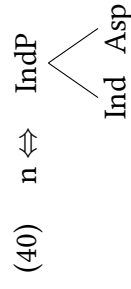
LIs for ThCh derivation



/-s/ IS LOST



/d-/ IS LOST



5.4 Diachronic account

Two big morphosyntactic changes occur in all modern Tibetic varieties, even though I only discuss ThCh here: a) the historical FUT stem is lost, and b) the tense-based distinction between the historical PRS and PST stems become an aspect-based distinction between IPV and PFV stems. In this section, I give an overview of these diachronic developments as localised to the CTib vs ThCh paradigms of $\sqrt{\text{kab}}$, modelled using NS constructs. The effects of historical phonology are pointed out when relevant.¹⁴ modelled using NS constructs. The effects of historical phonology are pointed out when relevant.

(42-43) repeats (19-20), with the showing the stems of $\sqrt{\text{kab}}$, relevant features, and exponents.

(42)

	SR	UR	stem	VP	Asp	End	Mood	Ind	Tense	Past	Addr
a)	<i>k^hobs</i>	o-kab-s	IMP	kab			-s				o-
b)	<i>bkab(s)</i>	b-kab-s	PST	kab			-s		b-		
c)	<i>ⁿgeb(s)</i>	n-V-kab	PRS	kab			V-		n-		
d)	<i>dgab</i>	d-V-kab	FUT	kab			V-		d-		

(43)

	SR	UR	stem	VP	Asp	End	Mood	Ind	Tense	Past	Addr
a)	[k ^h op]	o-kab	IMP	kab			o-				o-
b)	[kwap]	b-kab	PFV	kab			b-				
c)	[ŋgep]	n-V-kab	IPV	kab			V-	n-			

The change from a tense- to an aspect-based paradigm can be modelled as changes to the FSEQ features that are present in the paradigm. Some exponents that are inherited from CTib into ThCh change the features that they expone.

I present here a speculative sequence of developments:

1. Syllable-structure simplification from the historical phonology erodes /-s/ from CTib. The functions of /-s/ (Mood) is thus reattributed to /o-/ and /b-/ in IMP and PST respectively through reanalysis.
2. The use of the CTib PST stem in perfective contexts leads speakers to posit that the stem in fact has a perfective meaning, and thereby adding a second aspect feature to convey event boundedness (End). This creates a more developed aspect domain. In CTib, stems only contrasted on whether they had an aspect domain (no for IMP, yes for other stems), but because there were no additional features in the aspect domain, no aspectual distinc-

¹⁴ To overview other diachronic factors not under discussion here: The IMP stem is inherited without major change to its synsem content. All inherited stems are affected – to different degrees – by general phonological changes, namely syllable structure reduction, change in vowel qualities, tonogenesis in innovative varieties (Bielmeier et al., 2018 for systems across modern Tibetic; Haller, 2004 for ThCh). Individual verbs may be affected by analogical changes (paradigm levelling and extension (discussed in Liu, 2024). The ThCh stems of $\sqrt{\text{kab}}$ are all regular reflexes of its CTib stems (Haller, 2004).

tions were possible. Having two aspect features (Asp + Ind) produces a more specified aspectual interpretation (the event is bounded, i.e. perfective; after Wiland, 2025) than only having one feature (Asp only); the event is unbounded, i.e. imperfective).

3. This new aspectual distinction is somehow more salient or productive than the pre-existing tense distinction, so feature contrasts in the tense domain are eroded (losing Past), until the entire tense domain is eroded (losing Tense). We must posit the loss of the entire tense domain (losing both Past and Tense) rather than losing tense distinctions but not the domain (losing Past but not Tense) because the new aspectual stems are underspecified for tense, with temporal meanings being expressed periphrastically (Haller, 2004). The steps so far have created (43a-c), as well as a reflex of (42d) that lacks the Tense feature.
4. The CTib FUT stem already occurred less often than the PRS and PST stems, and its form less easily predictable (Liu, 2024). So with the emerging aspectual distinction, the more frequent CTib PRS stem becomes the ThCh IPV stem, and the CTib FUT stem merges into it. This results in the IMP : PFV : IPV paradigm in ThCh

This raises further questions for the motivations of these developments. One is why the aspectual distinction should be more salient or productive than the tense-based one. This could be due to usage – for example, if speakers gradually use the PST/PFV stem for non-durative events more than for durative ones, or if speakers periphrastically state tense information more alongside their use of the stem. How to investigate these usage effects, and what kinds of corpora may serve as a proxy for investigating historical frequency effects, are questions for further research.

Another possible avenue of future research is to ask whether NS makes specific predictions about which exponents are most easily eroded. One prediction may be that exponents which expone fewer feature are more easily eroded, and this may contribute to the loss of /-s/ (lexicalising Mood) or /d-/ (lexicalising Tense), in a mutually-reinforcing relationship with the phonological erosion of syllable structure. Another prediction might be that oppositions that are generated (i.e. made higher in the FSEQ) are more easily eroded if the stems in question are already distinguished lower down, making the aspectual distinction with End between more salient than the higher tense distinction.

6 Allomorphy and affix blocking: Difficulties and implications from generating CTib /s-/ and /V-/

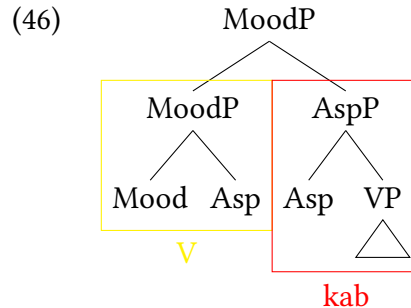
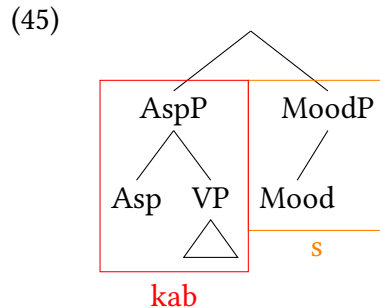
All the ThCh stems of $\sqrt{\text{kab}}$, and the CTib stems that do not have /V-/, can all be straightforwardly derived using the lexicalisation algorithm in (10) and the LIs presented above. In this section, I elaborate on my attempts to fix the derivation, from least to most successful. I begin by arguing that it makes sense to see /V-/ and /-s/ as allomorphs for the Mood feature, and that NS is promising for modelling allomorphy. Since there is no difficulty deriving the ThCh paradigm using my LIs and algorithm above, I focus on deriving the CTib paradigm here. (44) shows the CTib paradigm for $\sqrt{\text{kab}}$ again, repeating the table in (19) but without a greyed-out space for the End feature that is not present in CTib.

(44)	SR	UR	stem	VP	Asp	Mood	Ind	Tense	Past	Addr
a)	<i>k^hobs</i>	o-kab-s	IMP	kab		-s				o-
b)	<i>bkab(s)</i>	b-kab-s	PST	kab		-s		b-		
c)	<i>ⁿgeb(s)</i>	n-V-kab	PRS	kab		V-		n-		
d)	<i>dgab</i>	d-V-kab	FUT	kab		V-		d-		

6.1 Allomorphy of /-s/ and /V-/ in NS

Independent considerations mean that it makes sense to analyse /-s/ and /V-/ as allomorphs that both realise Mood. Firstly, /-s/ and /V-/ are in complementary distribution in the CTib paradigm (IMP and PST have /-s/, PRS and FUT have /V-/), but this distribution does not seem to reflect any featural natural classes, i.e. their distribution is ‘morphomic’ after Aronoff (1994). PRS and FUT do become a natural class when they both receive the Tense feature but not Past, but PRS and FUT or IMP and PST never become natural classes through actively sharing a feature. Secondly, both /-s/ and /V-/ also need to be derived at (or around) Mood for independent reasons. /-s/ must be at Mood because it is shared between IMP and PST, and Mood is the only feature they share other than those that comprise the verb root. /V-/ must be derived low because it is possible to have Tibetic verbs that take the other exponents in (19-20) but not voicing alternation. The most natural way to account for this in NS would be through differing root sizes: non-voice-alternating verbs have a bigger $\sqrt{\text{LI}}$ that can lexicalise the features that are covered by $\sqrt{\text{V}}$ and /V-/ in voice-alternating verbs. Thus /V-/ should be derived closest to the root to facilitate this. The possible options in my FSEQ here would be to say /V-/ is either derived at Asp or Mood. Considering that most stems have the Asp feature, it feels parsimonious to say that Asp is realised by the root, thereby putting /V-/ at Mood. Additionally and as discussed earlier, it is preferable for /V/ to be a prefix because then it can condition voicing in an immediately adjacent obstruent. These factors put /-s/ and /V-/ into an allomorphic relationship, where /-s/ is a suffixal realisation of Mood, and /V-/ a prefixal one.

The NS lexicalisation algorithm allows the same feature to be expressed in different positions in the derivation, as part of different hierarchical structures which correspond to different LIs. Thus it would be neat in NS to say that /-s/ and /V-/ spell out different positions of the Mood feature, derived through different lexicalisation-driven movements. /-s/ in (45) is derived when the root /kab/ is moved up and around Mood, and /V-/ in (46) is derived when Mood forms a CLB (the boxes in (45-46) respectively show the complete LIs for /-s/ and /V-/). This also means that, now, any issues around deriving one of /V-/ or /-s/ becomes an issue around deriving both of them.

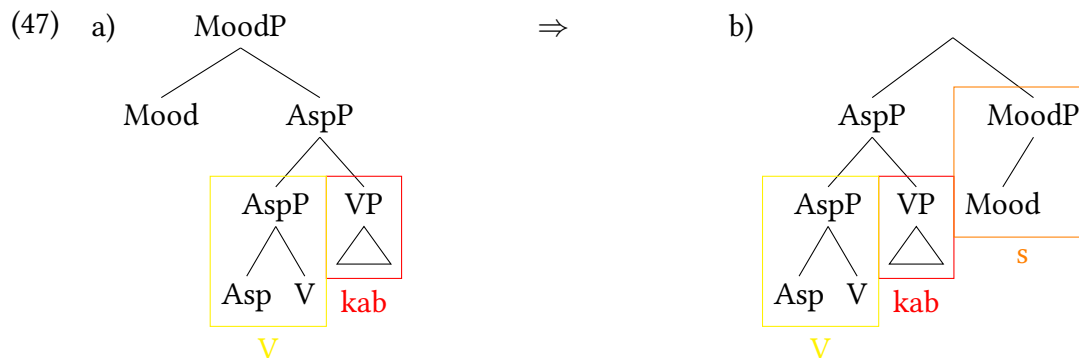


This is an elegant way of modelling allomorphy, except for the issue that, with the algorithm presented so far, it is impossible to construct a derivation that is able to make use of both of these LIs. Since both LIs lexicalise Mood, both should be derived in the Mood merge cycle. But /-s/ is derived from an earlier step in the lexicalisation algorithm (first movement), and because /-s/ can lexicalise successfully, the later step that derives /V-/ (CLB-formation) will never be triggered. Thus, the two stems that should have /V-/ will derive as */n-kab-s/ (PRS) and */d-kab-s/ (FUT) instead of the desired forms.

The fundamental thing that makes deriving the full CTib paradigm difficult is that the allomorphy of /-s/ and /V-/ is morphomic. If there was a feature that the voiced stems shared to the exclusion of the voiceless stems or vice versa, it would go far to dissolve the difficulties here. However, in the absence of such a feature (as per my discussion on synsem decompositions above, there is no natural candidate for such a feature), we must explore other routes to save the derivation. Some of the following options can successfully derive part of the paradigm but fail for others (albeit in a different way from the above), while other options can derive the whole paradigm but I reserve some level of scepticism for them. I present these options below from least to most promising.

6.2 Attempt 1: Derive /V-/ before /-s/

If, in the derivation presented so far, /-s/ blocks /V-/ by being the result of an earlier spellout option in the same lexicalisation cycle, it's natural to ask if we could simply derive /V-/ in the cycle before /-s/. Since /-s/ is fixed at Mood, we will use /-s/ $\Leftrightarrow$ [MoodP Mood] again but /V-/ $\Leftrightarrow$ [MoodP Mood [AspP Asp V]]. This means that when Mood is first merged we get (47a), and after movement we can – and can only – lexicalise as (47b). Since no stem has both /V-/ and /-s/, this derivation unilaterally fails.



6.3 Attempt 2: Overwrite

A more involved approach, which (like Attempt 1) keeps the same FSEQ and does not use additional technologies like backtracking or suppletion, would be to alter the relevant LIs more substantively, and ask if we can derive one of /-s/ or /V-/ first, and then overwrite it when the other one is merged. There are three options here: a) to overwrite the existing affix (/V-/ or /-s/) with the other one (/V-/ or /-s/); b) to overwrite one of /-s/ or /V-/ with later prefixes; c) to overwrite the existing affix (/V-/ or /-s/) with the root when the new feature/affix is merged.

As I argue below, none of these are possible under current NS technology, and they also reveal interesting implications of the lexicalisation algorithm.

Overwriting one affix with another. Here, the goal is to derive one of /V-/ or /-s/ first, and then, with the merging of the next feature, so the LI for /-s/ or /V-/ (the other affix) can lexicalise both the new feature and the features of the pre-existing affix. (48a) shows the minimal illustrative structure of a new feature Z being merged when we have already derived /V-kab/, and we would like to move the root (XP) to the left of the other features, so Z can form a constituent with the lower YP. We can then make /-s/ spell out the resulting constituent. There is no movement that would create this constituent. The first option for lexicalisation-driven movement would target the highest labelled non-remnant node, which is the higher YP. (48b) shows the same situation but vice versa: /kab-s/ has already been derived, and we need Z and YP to form a constituent to be lexicalised by /V-/. This is again not possible, because no movement would move Y or YP leftwards without XP.



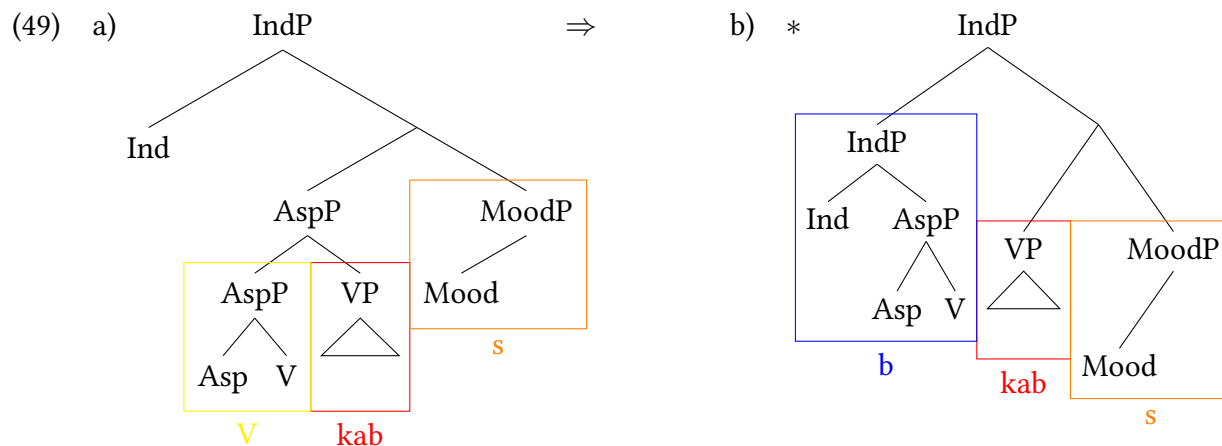
The possible lexicalisation outcomes for [prefix root] upon the merging of a new feature are [prefix root] or [prefix root suffix], never [root suffix].¹⁵ Same for vice versa. It is not clear to me whether this is a reasonable constraint in the NS lexicalisation algorithm, or simply a fall-out of how the existing algorithm has been constructed. While NS does commit strongly to an antisymmetric stance of prefixes vs suffixes, this distinction feels well-expressed by the distinction between unary- vs binary-bottom affixes, without extending it to predictions about overwrite during lexicalisation.

Overwrite /V-/ with /b-/. If affixes on different sides of the root cannot overwrite each other, maybe we can use overwrite on the same side of the root. /-s/ is the only suffix in the paradigm, but the paradigm is rife with different prefixes, so maybe we can try to overwrite /V-/ with a higher prefix when /V-/ is not needed.

As demonstrated above, because /-s/ must be derived at Mood, the only way to get both /V-/ and /-s/ in a derivation is as in Attempt 1, where /V-/ is generated at Asp and then /-s/ at Mood. Again as demonstrated, this means that all three stems that have both Asp and Mood will wrongly generate both affixes. Since prefixes cannot overwrite /-s/ in the stems where we do not want /-s/, our current approach already fails to correctly derive PRS and FUT. However,

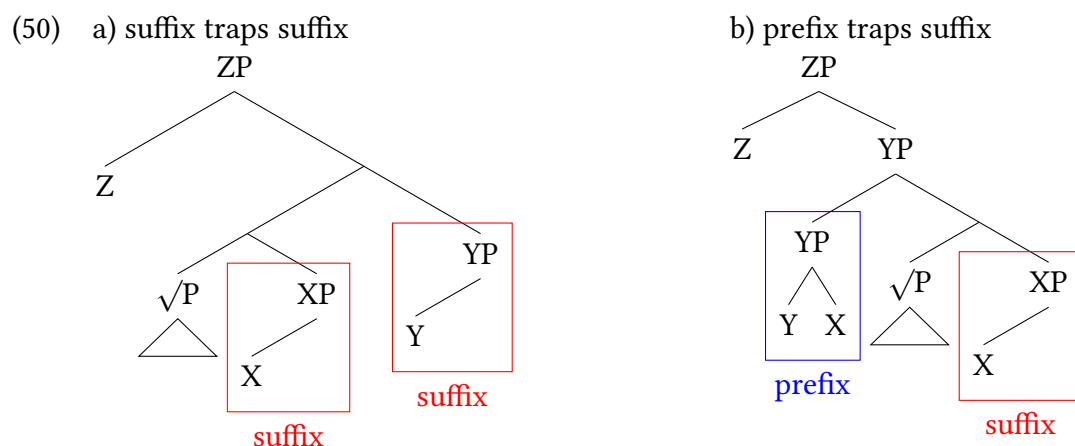
¹⁵ It is maybe possible to get root-suffix by changing the LI of the root – I discuss this possibility in the main text below.

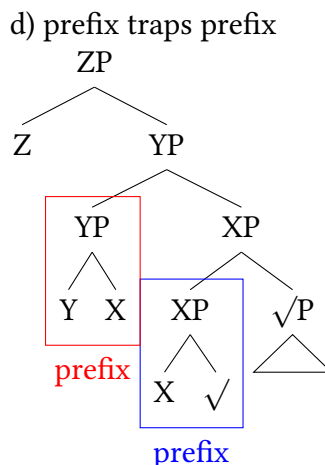
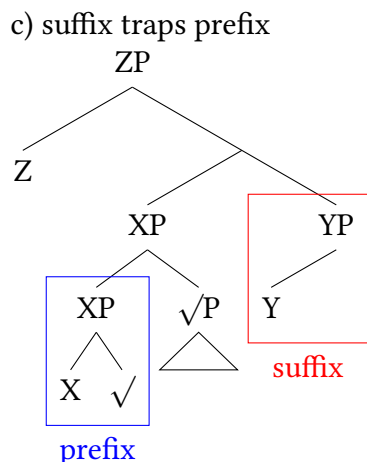
it's interesting to consider whether our current approach can successfully derive PST /b-kab-s/, which has Asp (so will generate /V-/) but cannot have /V-/ in the finished derivation. Here, the derivation generates */V-kab-s/ after Mood is merged, and when Ind is merged (49a), the LI for /b-/ overwrites /V-/ to get /b-kab-s/ (49b).



This turns out to also be impossible, as indicated by the * in (49b). Even though /b-/ and /V-/ are on the same side of the root, the features/projections that are first lexicalised by /V-/ (the lower AspP in (49a)) become inaccessible when /-s/ is derived above it, so no movement can make them form one constituent with the newly merged feature (Ind) to be lexicalised by /b-/.

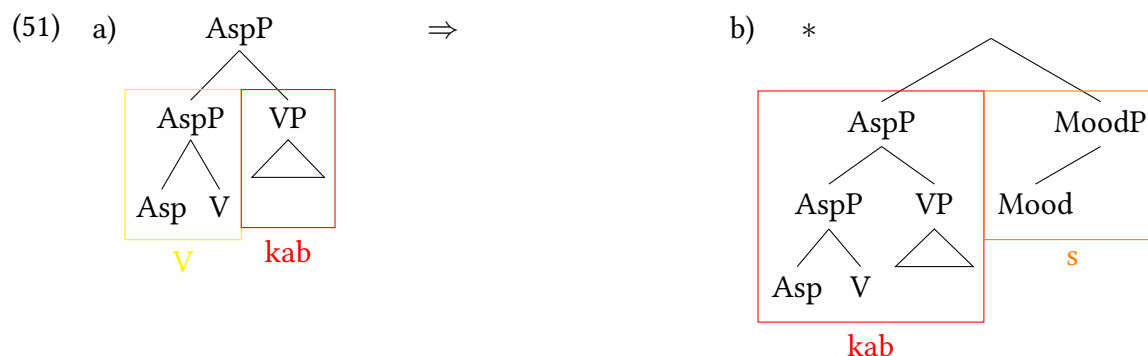
This is again part of a generalised prohibition in NS: affixes ‘trap’ the affixes below them. This holds for any combination of prefix and suffix, shown in (50). Only the features lexicalised by the most recently derived affix are accessible to future steps in the derivation. Features/projections lexicalised by a lower affix (XP) cannot move to form a constituent with a newly merged feature (Z), to the exclusion of other morphemes in the derivation ($\sqrt{P}$, YP).





Again, I don't know if this restriction is a deliberately intended part of the current NS lexicalisation algorithm, or if it falls out of how the algorithm happens to have been developed.

Overwrite the first affix with the root. In this approach, the full LI entry for the verb root is bigger and more complex, so it is able to overwrite the affix that we don't want to surface.¹⁶ The desired outcome is shown in (51b), where the full LI of the root turns out to be the big AspP, so can lexicalise the features that are lexicalised by /V-/ in (51a). This approach fails outright, because the root LI should already have lexicalised the full structure in (51a), meaning that the /V-/ LI would never have surfaced.



The three overwrite-based solutions surveyed in this section unilaterally fail, at least using the current NS derivational apparatus. My observations that prefixes and suffixes cannot overwrite each other, and that later affixes trap earlier affixes, are of interest for the general NS literature, to see whether these derivational constraints generate correct predictions in other case studies, and thereby whether these constraints should be retained in the algorithm.

¹⁶ Again like previously, to derive both /V-/ and /-s/, /V-/ must be derived first at Asp, and then /-s/ at Mood.

6.4 Attempt 3: Backtracking

Despite my scepticism against backtracking as voiced in Section 4, it is worth considering whether it would help us here. Here I use the LIs from Section 5.3, and backtracking as a final resort after CLB-formation fails (contra the standard literature where backtracking to precede CLB-formation). This method can derive the FUT stem successfully, but not PRS.

To derive the FUT stem (44d), we first derive /kab-s/ for the features up to Mood, as in Section 5.3. Tense is then merged, opening a CLB with Tense and Mood at the bottom. But there is no LI that has a binary bottom with Tense and Mood, so the derivation backtracks to the previous cycle, where we successfully lexicalised Mood by moving its complement out into spec position, as in (45). Because this structure produces no lexicalisable structures downstream, the next lexicalisation viable option from (10) is tried in the Mood cycle, which is to create a CLB with Mood and Asp. This creates the structure in (46), which then allows the /d-/ LI (34) to be derived on top of it in the next cycle, successfully lexicalising the FUT stem.

However, even with the inclusion of backtracking, this does not help the PRS stem (44c). Because its features are identical to those of PST (44b) until Past, PRS lexicalises with the same LIs as the PST stem, and backtracking is never triggered. Additionally, as per Section 4, the inclusion of backtracking at all is undesirable.

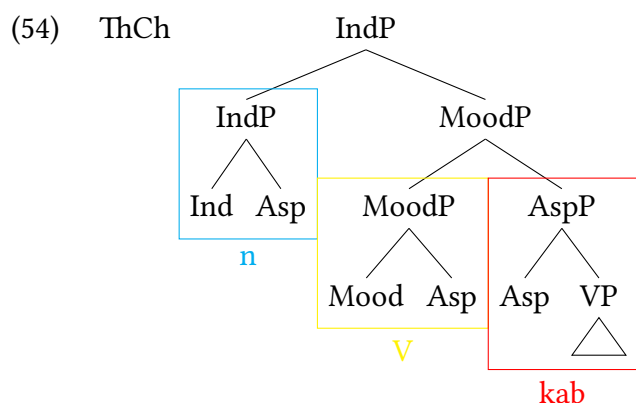
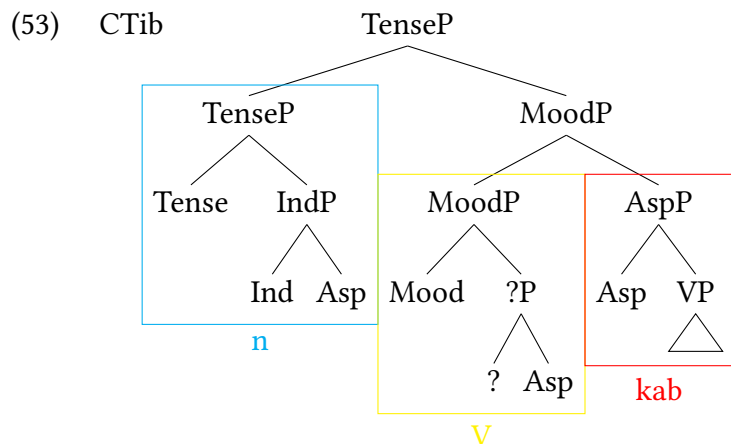
6.5 Attempt 4: Postulate null feature

The derivation of /-s/ and /V-/ in the correct places is made difficult by the fact that their appearance is not conditioned by a single featural difference. Therefore, we can trivially circumvent this issue by proposing a single conditioning feature that one of the LI has and the other doesn't. Since there is no good candidate for a semantically meaningful feature to demarcate /V-/ vs /-s/ stems, one can propose a semantically and syntactically null feature that exists solely to condition the allomorphic split. One possible place to insert this null feature is between Asp and Mood, shown as '?' in (52). The /V-/ LI with '?' is shown earlier in (31) in Section 5.3. Because stems with /-s/ do not have this '?' feature, they would not derive /V-/, and only derive /-s/ when Mood is merged.

(52)

	SR	UR	stem	VP	Asp	?	Mood	Ind	Tense	Past	Addr
a)	<i>k^hobs</i>	o-kab-s	IMP	kab			-s				o-
b)	<i>bkab(s)</i>	b-kab-s	PST	kab			-s		b-		
c)	<i>ngeb(s)</i>	n-V-kab	PRS	kab			V-		n-		
d)	<i>dgab</i>	d-V-kab	FUT	kab			V-		d-		

Thus, the full derivation for CTib PRS is (53). The split between /-s/ and /V-/ is no longer morphomic in ThCh (since 1PV is the only stem to have /V-/), so the '?' feature is lost in ThCh (54).



The obvious drawback of this analysis is that it involves positing a feature whose sole purpose is to condition allomorphy and is otherwise null. It is additionally unclear what the status of this feature would be in a cartographic landscape: is it also a feature on the universal spine, or some other kind of theoretical object? However, it also has the potential to be a powerful tool for deriving allomorphy if, in a system-wide analysis of CTib verbal morphology, it turned out that this same ‘?’ feature crucially helps to derive other morphomic allomorphies in CTib verbs.

6.6 Attempt 5: Suppletion

One native NS strategy for suppletion is the use of ‘pointers’, which are LIs that reference other LIs (e.g. *better* ⇔ [*good* Comparative]). For example, a separate LI /gab/ would overwrite /kab-s/ in the presence of /n-/ and /d-/. Voice-alternating verbs have this additional root LI, non-voice-alternating verbs do not.

I hesitate to fully endorse the suppletive analysis because of the morphophonological regularity of the verb stem voicing alternation. Suppletion definitely has a place in the CTib verbal system, for modelling paradigms with non-cognate forms (e.g. CTib ‘to go’ has PRS stem ⁿ*gro* and PST stem *song*), and it feels overly convenient to use it also to model a morphophonologically regular process. However, since there are only 50 voice-alternating verbs in Hill (2010), it is not prohibitive to propose that this relatively small class of verbs should have an extra supple-

tive root LL. The strength of having a pointer that introduces the voiced root in the presence of /n-/ or /d-/ is that it models the fact that, for all 50 voice-alternating verbs in Hill (2010), their voiced stems always occur with /n-/ or /d-/¹⁷ (but stems with /n-/ or /d-/ are not always voiced).

7 Discussion: Achievements and limitations

The analysis in Section 5 shows that it is possible to model instances of opaque GE-style exponence in Tibetic using formal decompositional frameworks. Specifically, it is possible to give morphological phenomena with many-to-many form-function mappings a decompositional analysis if we admit an intervening abstract layer between the ‘form’ and ‘function’ of traditional morphemes, the intervening layer here being the LIs and lexicalisation algorithm of NS. Even though the morphomic distribution of /-s/ vs /V-/ produces difficulties for the analysis, Section 6 shows that these difficulties can be surmounted either using existing NS technologies or perhaps by re-examining the current algorithm. Thus I think the analysis remains insightful overall despite those issues.

My own chief reservation about this analysis is the attribution of FSEQ features to phonological exponents. Which UR exponents are mapped to which FSEQ features in this analysis have been largely guided by considerations of matching feature-ordering in the FSEQ against the order in which exponents should be generated, and not through traditional structuralist comparison between the form and function of different stems. For example, /V-/ $\Leftrightarrow$ [_{MoodP} Mood Asp] in this analysis means ‘the best way to draw our trees is for /V-/ to correspond to the (Asp-)Mood section of the cartographic spine’, and not ‘/V-/ seems to express a modal (and aspectual) function because its appearance correlates with this synsem interpretation’. Viewed positively, this kind of maneuvering is a natural result of modelling opaque form-function mappings, and exemplifies how theory can be used to constrain linguistic hypotheses, but viewed negatively, one could argue that this amounts to assigning functions to forms by brute force, out of thin air. One could argue that this is not a problem as long as one is always clear about which claims have or have not been stipulated using theory-internal means, but there is still the possibility of confusion and misconstrual. This is perhaps another problem for the consideration of future work.

To my knowledge, this is the first time formal decomposition has been attempted for Tibetic verbal morphology (esp. CTib), and also the first time NS has been applied to Tibetic languages. Even though this study is limited in scope, the overall success of using NS to generate the paradigm for $\sqrt{\text{kab}}$ provides proof of concept that this approach is fruitful, and could be expanded to model Tibetic verbal systems more generally. That project would have further difficulties – notably, navigating complex and ill-understood morphological classes (cf. Liu, 2024) and disagreements in the field about the synsem interpretation of CTib verb stems (Zeisler, 2004 vs Bialek, 2020) – but it would follow the theoretical groundwork laid out here.

¹⁷ I ran regex searches in a plain-text version of the Hill (2010) verb dictionary to verify this. This observation is interesting per se, but difficult to interpret without further research into the etymology of those verbs specifically (e.g. in the style of Hill, 2005), since verbs with marked morphology are also likely to experience prescriptive and/or subconscious pressures towards regularisation.

By /n-/ I mean *n*-, and by /d-/ I mean *d*- and its allomorph *g*-.

This study also represents an early attempt to bring NS into diachronic investigations (following e.g. De Clercq, 2020). The NS account here produces a descriptively successful model of the proposed diachronic developments in Section 5.4, which are ultimately driven by the cognitive process of reanalysis. On the one hand, this means that my diachronic account does not require NS to be coherent, but on the other hand, NS does not appear to add significant explanatory value to the changes proposed here. Because diachronic research using NS is still relatively underexplored compared to synchronic NS and other diachronic generative approaches, it is difficult to determine at present whether the diachronic NS account presented here strikes a satisfactory balance between descriptive and explanatory adequacy. Additionally, the changes proposed here are speculative, and their veracity and periodisation will have to be determined with future research using comparative evidence and cognitive modelling of language change.

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